

Pollinator Ecology and Management

Chicago Bees: Urban Areas Support Diverse Bee Communities but With More Non-Native Bee Species Compared to Suburban Areas

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Abstract

Urbanization is rapidly growing worldwide, yet we still do not fully understand how it affects many organisms. This may be especially true for wild bees that require specific nesting and floral resources and have been threatened by habitat loss. Our study explores the response of wild bee communities to an urbanization gradient in the Chicagoland region of Illinois. Specifically, we explored how both landscape scale impervious surface and local floral diversity across an urbanization gradient influenced 1) the composition of local bee communities, 2) the richness of native and non-native bees, and 3) the composition of bee functional traits. Over the course of our study, we documented 2,331 bees belonging to 83 different species, 13 of which were not native to North America. We found that impervious surface influenced the overall composition of bee communities. In particular, highly urban areas were composed of more non-native bee species and fewer native bee species. Additionally, bee richness and native bee richness responded positively to floral resources. Bee functional trait responses were variable, with floral diverse sites supporting greater richness of ground nesting, eusocial, and generalist bees regardless of landscape-level impervious surface. Importantly, our study provides evidence that urban areas can support diverse bee communities, but urban and suburban bee communities do differ in composition. Thus, bee conservation efforts in urban areas should focus on creating floral diverse habitats to help support more bee species, specifically native bee species, while also considering which bees are best supported by these conservation efforts.

Key words: bees, Chicago, community ecology, functional trait, pollinator

Urbanization is a human driven land-use change that is rapidly increasing worldwide, posing a threat to many natural ecosystems (Czech et al. 2000, McKinney 2002, 2006). Urban areas are defined by impervious surfaces (e.g., pavement and structures), with disconnected patches of vegetation, both of which can affect the local persistence of species. It has been estimated that in highly urban areas as little as 20% of land cover is various forms of vegetation or ‘greenspace’ (e.g., managed parks, unmanaged vegetation; Sukopp and Werner 1982). Evidence suggests that urbanization generally homogenizes plant and animal communities and reduces native species abundance and richness (McKinney 2006, 2008). As urban areas continue to increase globally, more and more species are forced to interact with and inhabit these environments. Therefore, understanding how species and communities respond to urbanization is

crucial, yet we are only beginning to understand the consequences of urbanization on biodiversity (McKinney 2008, Winfree et al. 2011).

Bees (Hymenoptera: Apiformes) make up 20,000–30,000 species worldwide (Michener 2007) and are ecologically and economically important (Winfree et al. 2007, Potts et al. 2010, Garibaldi et al. 2013). However, many bee species have been experiencing recent declines (Biesmeijer et al. 2006, Potts et al. 2010, Burkle et al. 2013, Goulson et al. 2015), in part due to land-use change—including urbanization (Winfree et al. 2011, Cardoso and Gonçalves 2018). For example, in the city of Curitiba, Brazil, bees declined in abundance by 45% over 34 yr in response to increased urbanization (Cardoso and Gonçalves 2018). The reduction of bee species richness and diversity in urban settings is generally attributed to increases in impervious surface and losses

of greenspace. Greenspaces provide bees with places to live (nesting habitat) and food (floral resources), thus the lack of greenspaces in urban settings may potentially limit bee populations in cities (Roulston and Goodell 2011, Winfree et al. 2011). Although urbanization has been considered a threat to bees, it has also recently been argued that urban areas may be havens for bee diversity, in part owing to managed gardens or other greenspaces that provide floral and nesting resources for bees (Hall et al. 2017).

Indeed, some studies that have investigated the effects of urbanization on wild bee communities have reported that urbanization can have strong negative effects on bee communities, whereas others find strong positive effects (Supp Table S1 [online only]). Furthermore, several studies have found that within urban areas, local floral resources can act as important positive drivers of bee diversity (Matteson and Langellotto 2010, Pardee and Philpott 2014, Lowe and Foltz-Sweet 2017, Choate et al. 2018, Wilson and Jamieson 2019). The availability of resources (food and nesting) in urban areas may compensate for the negative effects of urbanization and might help to explain the variable results across studies (Supp Table S1 [online only]). If urban areas have sufficient resources, bee diversity and abundance may be maintained. Alternatively, if urban areas are poor in resources, bee diversity and abundance may decline as a result. Thus, it is essential to understand how both landscape-scale urbanization (impervious surface) and local landscape features (e.g., floral resources) affect bee communities in urban areas. Furthermore, it is critical to examine the identities of the bee species supported by urban areas. For example, urban areas may harbor abundant and diverse bee communities, but it is possible that these communities may be dominated by non-native species capable of persisting in highly modified and disturbed habitats (Matteson et al. 2008, Fitch et al. 2019, Wilson and Jamieson 2019).

In addition to bee abundance and diversity, the composition and traits of bees may differ between urban and nonurban areas. Urbanization may act as a strong ecological filter, resulting in high turnover in bee community composition along a gradient of urbanization. This turnover may be related to specific traits that may mediate whether or not the bee is able to persist in urban environments (Wenzel et al. 2020). For example, dietary specialists may be less common in urban areas if their preferred floral resources are in low abundance or not present. Yet, the extent to which bee functional traits allow bees to persist in urban environments remains unclear. Although several studies have reported a higher abundance of above-ground nesting species in highly urban areas (Supp Table S2 [online only]; Matteson et al. 2008, Fortel et al. 2014, Cardoso and Gonçalves 2018), other bee functional traits, including as sociality, diet breadth, tongue length, and body size, have yielded variable results (Supp Table S2 [online only]).

Furthermore, investigating communities along an environmental gradient can help form a predictive backbone for the structure of ecological communities (McGill et al. 2006). Yet, this approach has only recently been used to explore ecological communities in urban landscapes (Banaszak-Cibicka and Żmihorski 2012, Fortel et al. 2014). By investigating bee communities along an urbanization gradient, we can identify if urbanization acts as a filter, resulting in turnover in bee community composition.

In this study, we examined wild bee communities along an urbanization gradient spanning from the urban core of the City of Chicago out into its northern suburbs. Here, we investigated how landscape-scale urbanization (impervious surface) and local floral resources along this urbanization gradient affect 1) bee species richness, abundance, diversity, and composition, 2) the richness of native and non-native bees, and 3) the composition of bee functional traits.

Materials and Methods

Study Sites

The study was conducted in 2018 at eight sites in the greater metropolitan area of Chicago, IL, United States (Cook county) located along the Union Pacific North Metra (UP-N) railway which runs from the urban core of Chicago into the northern suburbs (Fig. 1). All eight sites were directly adjacent to the UP-N train stations. Five sites surrounded the tarmacs of UP-N train stations (Clybourn, Ravenswood, Rogers Park, Evanston Main, and Evanston Central), and three sites were located along the Green Bay Trail (a pedestrian path that sits right next to the UP-N rail line), all of which were next to train stations (Indian Hill, Hubbard Woods, and Braeside) (Fig. 1). Permission to sample at sites was provided by the City of Chicago (Clybourn, Ravenswood, Rogers Park), the City of Evanston (Evanston Main and Evanston Central), and Friends of the Green Bay Trail (Indian Hill, Hubbard Woods, Braeside). The distance between adjacent sites along the urbanization gradient ranged from 2.7 to 5.5 km. We conducted bee and plant community surveys at each site four times throughout the growing season (once in June, twice in July, and once in August). Sampling occurred on warm sunny days for 8-hour periods, generally beginning at 0830 and ending at 1630. Weather conditions (temperature, wind speed, and humidity) were measured at the beginning and end of each sampling period using a Kestrel 5700 pocket weather station (Kestrel Meters, Boothwyn, PA). While conducting bee and plant community surveys, we sampled one or two sites a day. Each site was paired with another site at least 11 km away, resulting in four site pairs. Each site pair was sampled on two consecutive days (or the next sunny day) to avoid sampling sites in close proximity on successive days. At each site, we established two 100 × 3 m transects on each side of the tarmac or trail to sample floral resources and bee communities. The only exception was for Ravenswood, which was sampled on only one side due to road construction. The plant communities surrounding each train station sites are minimally managed, and are relatively similar, being composed of many ruderal species (Supp Fig. S1 [online only]).

Quantifying Urbanization

To quantify the level of urbanization around each site (i.e., landscape-scale landscape features), we quantified land cover for the surrounding landscape within a 500 m and 1,000 m radius of each site using Geographic Information Systems (ArcGIS) and the 2011 National Land Cover Database (Multi-Resolution Land Characteristics Consortium, mrlc.gov). Land cover types included water, tree canopy, roads and railroad, other paved surfaces, grass and shrub, buildings, and bare soil. Buildings, road and railroad, and 'other paved surfaces' land cover types were used to determine the percent impervious surface surrounding each site (Table 1). Our study sites represented a clear gradient with the percent impervious surface surrounding each site increasing dramatically toward the urban center of Chicago (15–69% at 500 m radius and 13–71% at 1,000 m) and greenspace decreasing (85–26% at 500 m, and 90–35% at 1,000 m) (Fig. 1; Table 1).

Quantifying Floral Resources

To quantify floral resources available at each site and each sampling period, we divided the two 100 × 3 m transects into eight 25 × 3 m sections. During each sampling session, and in each section, we recorded the identities of flowering species and counted the number of

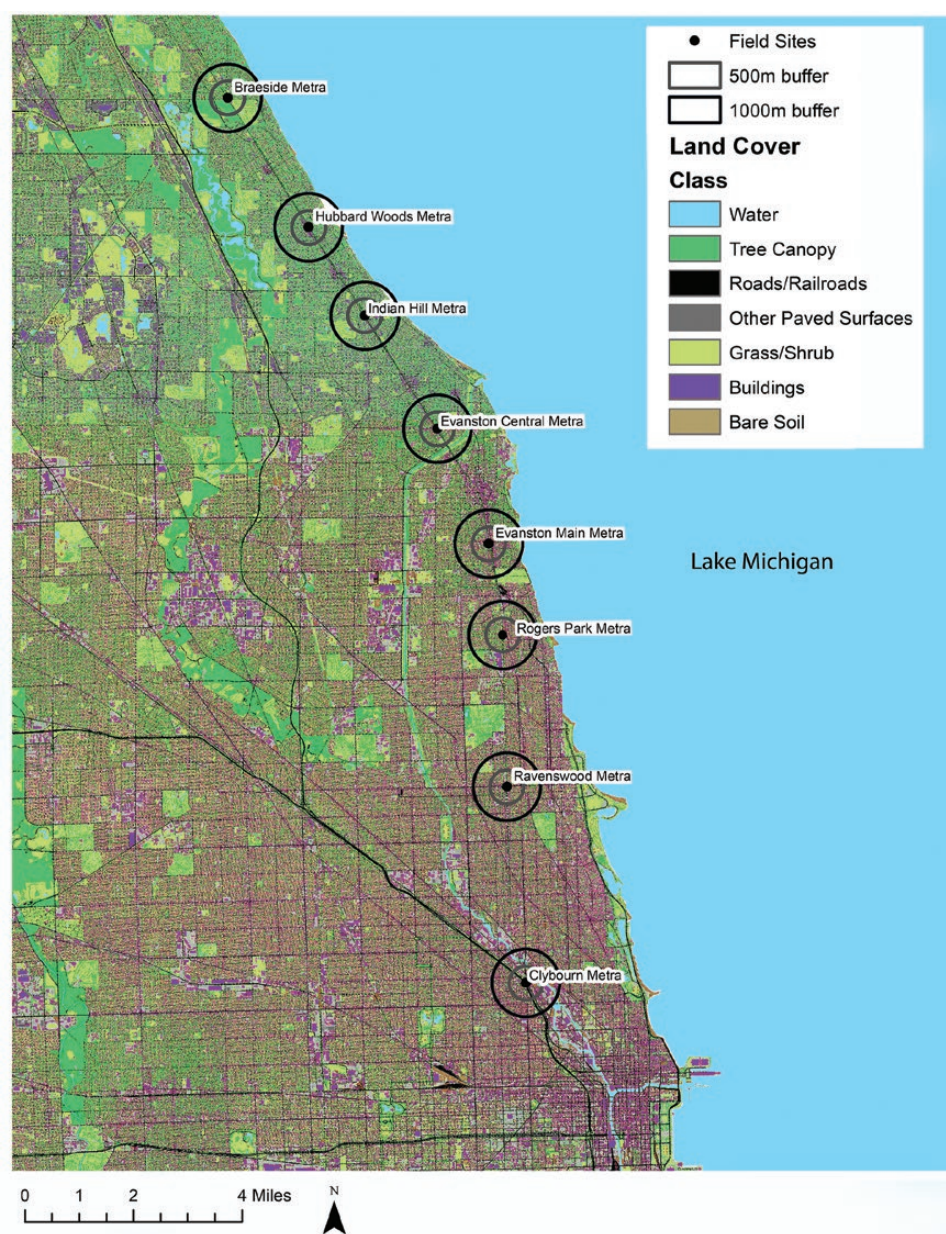


Fig. 1. Map of sampling area and sites showing impervious surfaces (roads/railroads, buildings, other paved surfaces) and green space (grass/shrub, tree canopy) in the Chicagoland region. Circles around sites represent 500 m and 1000 m radii that were used to calculate % impervious surface and % greenspace surrounding each site. Color figure is available online.

flowering inflorescences of each species. Some of our sites had very few species, but otherwise had a high abundance of inflorescences, in most cases due to non-native plant species. To account for this, we summarize floral resources at our sites by calculating local floral diversity (Shannon–Weiner diversity index) for each sampling period, using the abundance of inflorescences of each species.

Quantifying Bee Communities

To maximize the capture of diverse taxa, bees were collected using a combination of hand-netting and pan traps. We deployed 10 sets of pan traps during each sampling period; each set consisting of a blue, yellow, and white 3.25 oz plastic bowl as described by the USGS Bee Inventory and Monitoring Lab (BIMI; Droege 2015). All pan trap sets were placed along the 100 × 3 m transects at least 10 m apart and set on the ground unobscured by vegetation. Traps remained in

the field for the 8-hour sampling period. In addition to pan traps, we hand-netted bees within the 100 × 3 m transects at each site for 15 min every two hours over the 8-hour period (4-samples per period = 1 h of netting total per sample session). Sampling effort, including the number of pan traps and duration of netting sessions, was the same for every site (including Ravenswood, which had only one transect due to construction). All netted bees were stored in ethyl acetate-charged jars corresponding to the plant species on which they were collected. During some sampling periods, the abundance of honey bees was very high. In these instances, we ceased the collection of honey bees after two netting periods and walked along the transect counting honey bees to estimate abundance in order to prevent honey bees from dominating our sampling. All captured bees were transported to the laboratories at the Chicago Botanic Garden and dried and pinned within two days of collection.

Table 1. Attributes for all eight field sites including locale information and % impervious surface and greenspace at 500 m and 1,000 m radii

Attribute	Braeside	Hubbard Woods	Indian Hill	Evanston Central	Evanston Main	Rogers Park	Ravenswood	Clybourn
Distance from city center (m)	29.8	28.4	25.3	21.4	17.8	15.1	10.6	5.5
% impervious surface 500 m	15.1%	27.2%	30.9%	38.6%	57.3%	52.1%	61.6%	69.8%
% greenspace 500 m	84.5%	72.8%	69.0%	61.4%	42.7%	47.9%	38.4%	26.1%
% impervious surface 1,000 m	12.7%	22.1%	23.9%	29.6%	43.4%	52.5%	55.2%	71.2%
% greenspace 1,000 m	90.3%	80.7%	75.9%	70.9%	58.7%	47.5%	44.8%	35.1%
Latitude	42.1521	42.12159	42.0923	42.0627	42.0333	42.00925	41.9666	41.9172
Longitude	-87.7718	-87.7462	-87.7217	-87.6973	-87.6796	-87.6755	-87.6746	-87.668

All bees were identified to the lowest taxonomic level possible: 96% were identified to species, with the remaining bees identified to the genus or morphospecies level. Bees were identified using Discover Life (Ascher and Pickering 2020), taxonomic references (Mitchell 1962, Michener et al. 1994, Michener 2007), and a reference collection provided by the University of Illinois at Urbana-Champaign. For all species, a range of functional traits was compiled from the literature (Michener 2007, Williams et al. 2014), including: nesting behavior (ground or above ground nesting), sociality (eusocial, noneusocial or parasitic), and diet breadth (oligolectic or polylectic), as well as if the species is native or non-native to North America. Species in the genus *Lasioglossum* (subgenus *Dialictus*) were considered eusocial. In addition, we also estimated body size for each species by measuring the intertegular distance of 10 male and 10 female bees of each species using a Zeiss stemi 2000c stereomicroscope with a scale-bar (Carl Zeiss Microscopy, LLC, White Plains, NY). For species where we collected fewer than 10 males and 10 females, we measured all available specimens. We then calculated an average size for both males and females of each species.

Statistical Analyses

Relationships Among Predictor Variables

To better understand variation in our landscape and local-scale landscape predictors we used linear models to explore the relationship between percent impervious surface within both 500 m and 1,000 m radii surrounding each site as well as the relationship between percent impervious surface and floral diversity at the local scale (Shannon–Weiner diversity index). Percent impervious surface at the 500 m and 1,000 m radii was highly correlated ($R^2 = 0.92$, $P < 0.0001$), but we found no relationship between percent impervious surface and floral diversity ($R^2 = 0.061$, $P = 0.21$). Given the strong relationship between the percent impervious surface at scale of 500 m versus 1,000 m radii, we focused on the percent impervious surface at 500 m in all of our analyses for simplicity. To better understand the relationship between native and non-native floral resources, we explored the proportional abundance and richness of non-native inflorescences and their relationship with percent impervious surface (500 m) using generalized linear mixed effects models (GLMM) with a binomial distribution and site included as a random effect. Models were weighted by total floral abundance and total floral richness respectively.

- 1) *How does landscape-scale impervious surface and local floral resources influence wild bee species abundance, richness, diversity, and composition?*

To investigate bee community richness and sampling effort, we first categorized sites as suburban, with <50% impervious surface within a 500 m radius (Braeside, Indian Hill, Hubbard Woods, Evanston Central), or urban, sites >50% impervious surface within 500 m

radius (Evanston Main, Rogers Park, Ravenswood, Clybourn) (Table 1) and explored rank abundance distributions and species accumulation curves using BiodiversityR, vegan, and iNEXT packages in R (Kindt and Coe 2005, Hsieh et al. 2016, Oksanen et al. 2019). We calculated sampling completeness with the observed species richness and estimated expected asymptotic species richness using the Chao1 index. Next, we created a nonmetric multidimensional scaling (NMDS) ordination with Bray–Curtis dissimilarity index to compare bee species composition between urban and suburban sites using the vegan package (Oksanen et al. 2019). Ordinations were created treating each sampling period as a replicate and with three axes to minimize stress. We then used permutational multivariate analysis of variance (PERMANOVA) to test for differences between urban and suburban bee community composition (Anderson 2017). To further explore bee composition along the urbanization gradient, we used a linear mixed effects model (LMM) with Bray–Curtis dissimilarity values calculated for each site combination during each sampling month (June, Early July, Late July, and August) as a response variable, the change in percent impervious surface between sites (at 500 m) as a predictor, and sampling period as a random effect.

To explore the relationships between landscape variables and bee community metrics, we performed LMMs and GLMMs using the lme4 (Bates et al. 2015) and glmmTMB (Brooks et al. 2017) packages in R, respectively. The following community metrics were calculated for each sampling period at each site along the urbanization gradient and used as our response variables: bee abundance, bee richness, bee diversity (Shannon–Weiner diversity index), and bee evenness. We used two predictor variables in our models: percent impervious surface (500 m) and floral diversity (Shannon–Weiner diversity index). Five models were tested for each response variable: (i) a null model, (ii) a model including only percent impervious surface, (iii) a model including only floral diversity, (iv) a model including both percent impervious surface and floral diversity, and (v) a model including the interaction between percent impervious surface and floral diversity. For continuous response variables (bee diversity and evenness), we used LMMs. When response variables were count data (bee abundance and bee richness), we used GLMMs with a Poisson error distribution (or negative binomial distribution if overdispersion was detected). For all models, we included site as a random effect to account for sampling multiple times at one site throughout the season. The best fit models were selected using Akaike Information Criterion (AICc). The model with the lowest AICc value was considered the best fit model, and models <2ΔAICc values were considered as competing models. Type II ANOVAs were used to assess statistical significance, except when the best model included an interaction term, whereby we used a Type III ANOVA. An outlier was present in both bee evenness and bee diversity data, although removing this outlier did not affect model selection or results (not shown), and thus it was retained in all analyses.

2) *How does landscape-scale impervious surface and local floral resources influence native and non-native bee species?*

To investigate the relationship between native and non-native bees and our landscape variables, we first calculated the richness of both native and non-native bees at each sampling period at each site. We then used GLMMs with native and non-native bee richness as response variables and with percent impervious surface and floral diversity as predictors. The same set of models, model selection methods, and error distributions used to address our first question were used to determine the models that best supported the richness of native and non-native bees along the urbanization gradient.

3) *How does landscape-scale impervious surface and local floral resources influence bee functional traits?*

To assess the relationship between bee functional traits and landscape variables, we categorized all bees into the following functional trait groups: above-ground nesting or ground nesting; eusocial, noneusocial, or parasitic; polylectic or oligolectic. We calculated the richness of bees within each functional trait category for each sampling period at each site across the urbanization gradient. Given the low number of parasitic bees collected across our study, we examined the richness of parasitic bees collected at each site over the entire season (combining all sampling periods for each site). To investigate the relationship between functional traits and landscape variables we used GLMMs with the richness of bees with each functional trait as a response variable and percent impervious surface and floral diversity as predictors. We used the same models and model selection methods as described in previous sections (we used a GLM for parasitic bees because all sampling periods were combined). In addition to the categorical functional traits described above, we also explored the relationship between bee size (measured as the average intertegular distance) and landscape variables (percent impervious surface and floral diversity) with LMMs. The average size of bees was calculated for each sampling period at each site and used as a response variable in models.

Results

Flora

Across the urbanization gradient, we recorded 127 plant species (or morphospecies) representing 41 plant families, and an estimated 274,850 flowering inflorescences across the study period. Asteraceae accounted for the greatest number of species, 33 (27%), followed by 9 Fabaceae (8%), 9 Rosaceae (8%), and 6 Lamiaceae (5%). Non-native and ornamental species were the most abundant across all sites accounting for 241,430 (88%) inflorescences and 70 species (55%), whereas native species accounted for 33,420 (12%) inflorescences and 57 species (45%). The best models explaining the proportional abundance and richness of non-native inflorescences

included percent impervious, but both had a competing null model ($<2\Delta\text{AICc}$) (Table 2; Supp Table S5 [online only]). The abundance and richness of non-native inflorescences was the greatest at sites with a high percentage of surrounding impervious surface (Supp Fig. S2 [online only]).

Bee Fauna

Over the course of our sampling, we documented 2,331 bees, representing 83 species and five families: Andrenidae, Apidae, Colletidae, Halictidae, and Megachilidae (Supp Tables S3 and S4 [online only]). The greatest number of species were from the Megachilidae family, whereas we captured the most individuals from the Apidae family (Supp Table S3 [online only]). The most common bee documented along the urban gradient was *Apis mellifera* (L.) (Apidae), followed by *Megachile rotundata* (Fabricius) (Megachilidae), *Agapostemon virescens* (Fabricius) (Halictidae), and *Halictus ligatus* (Say) (Halictidae) (Supp Table S4 [online only]). We documented two critically endangered rusty patched bumble bees (*Bombus affinis* (Cresson)) at one of the urban sites (Rogers Park; these bees were not collected). We also documented a new Illinois state record: *Megachile apicalis* (Spinola) (Megachilidae), a non-native species, which was found exclusively at our most urban site (Clybourn; Gruver and CaraDonna 2020). The majority of bees were collected by hand-netting (1,739 bees, 80.6%) compared to pan traps (418, 19.4%). We collected 77 species through netting, 42 of which were unique to netting and not captured in pan traps. In contrast, pan traps captured 41 different species with six being unique to pan traps and not captured through netting.

Of the 83 bee species documented, 43 species (52%) nested in the ground while 40 species (48%) nested above ground. Within species that nested above ground, 28 species (34%) nested in cavities, followed by 4 species (5%) in pithy stems, 4 species (5%) in wood, 2 species (2%) constructed nests in the open, and 2 species (2%) nested in hives. True eusocial bees made up 20 species (24%), while 54 species (65%) were not eusocial, 8 species (10%) were parasitic, and 1 species (1%) had unknown sociality. Among the noneusocial bees, 46 species (55%) were solitary, 4 species (5%) sub-social, 3 (4%) solitary and eusocial, and 1 species (1%) solitary and communal. The majority of bee species were polylectic (74 species, 89%), whereas oligolectic bees accounted for only 8 species (10%); 1 species (1%) had unknown diet preferences. The intertegular distance of bees ranged from 0.90 to 5.38 mm (mean = 2.04 mm) for female bees and 0.84 to 4.66 mm (mean = 1.99) for male bees. Eleven species were represented by only a single individual (13% of species).

Finally, bees native to North America were the most abundant, accounting for 1,291 individuals (55%) and 70 (84%) species, whereas non-native bees, including honey bees, accounted for 1,040 individuals (44%) and 13 species (16%). Of the bees that were netted while foraging on flowers, 841 were native and 694 were non-native. Native bees were captured visiting native plants 23%

Table 2. Model selection results for the effects of % impervious surface (500 m) and floral diversity (Shannon–Weiner index) on the proportional richness and abundance of non-native inflorescences

Response	Model	K	ΔAICc	Relative likelihood	Weight
Proportional richness of non-native inflorescences	% impervious surface	2	0.00	1.00	0.53
	Null	1	0.27	0.87	0.47
Proportional abundance of non-native inflorescences	% impervious surface	2	0.00	1.00	0.66
	Null	0	1.36	0.51	0.34

The best fit models are in bold.

of the time, and non-native plants 77% of the time; non-native bees were captured visiting native plants 12% of the time, and non-native plants 88% of the time.

1) *How does landscape-scale impervious surface and local floral resources influence wild bee species abundance, richness, diversity and composition?*

Species richness accumulation curves indicated an adequate sampling effort at both urban and suburban sites (sampling completeness was approximately 94.5% for suburban sites and 93.8% for urban sites) (Supp Fig. S3a [online only]) with suburban sites being composed of overall more species than urban sites (70 vs 53; Supp Fig. S3 [online only]). NMDS ordination revealed significant compositional differences in bee communities between urban and suburban sites (PERMANOVA: $P < 0.01$) (Fig. 2). However, we did not find a relationship between variation in percent impervious surface between sites and Bray–Curtis dissimilarity (β diversity) between sites (Table 3; Supp Table S6 [online only]).

Bee species abundance, richness, evenness, and diversity did not respond consistently to landscape-scale impervious surface and local floral resources. Bee abundance was best explained by the null model, but had three additional competing models ($<2\Delta AIC_c$), the first including percent impervious surface ($\chi^2 = 1.16$, $P = 0.28$), the second including floral diversity ($\chi^2 = 1.12$, $P = 0.29$), and a third including percent impervious surface ($\chi^2 = 6.77$, $P = 0.009$), floral diversity ($\chi^2 = 7.19$, $P = 0.007$), and their interaction ($\chi^2 = 5.06$, $P = 0.025$). Bee species richness was best supported by a model including floral diversity, whereby bee richness increased with greater floral diversity (Table 3; Supp Table S6 [online only]; Fig. 3). Bee evenness and diversity was best supported by null models (Table 3; Supp Table S6 [online only]; Fig. 3).

2) *How does landscape-scale impervious surface and local floral resources influence native and non-native bee species?*

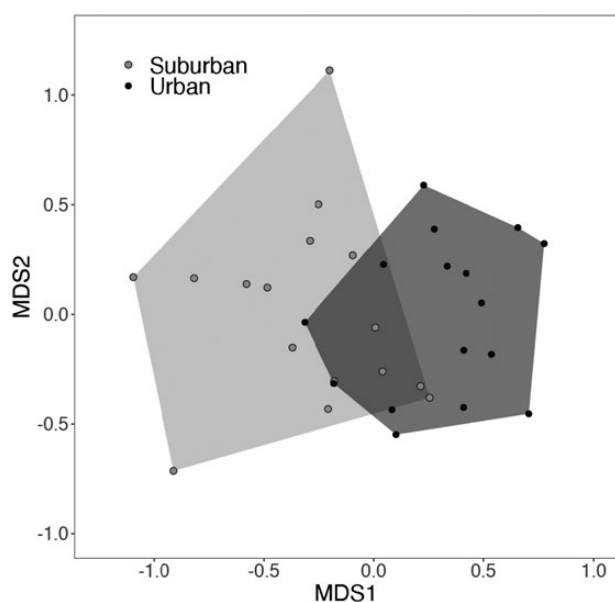


Fig. 2. Nonmetric multidimensional scaling (NMDS) ordination of bee species composition between suburban and urban sites of the urbanization gradient (stress = 0.17), dots represent sampling site, and ellipses represent habitat. Bee species composition differed between urban and suburban sites (PERMANOVA: $p < 0.01$).

Non-native bee richness was best supported by a model including percent impervious surface, whereby non-native bee richness increased at sites with a greater percentage of surrounding impervious surface (Table 4; Supp Table S7 [online only]; Fig. 4). Native bee species richness was best supported by a model including both percent impervious surface and floral diversity, whereby native bee richness increased at sites with a lower percentage of surrounding impervious surface and greater floral diversity (Table 4; Supp Table S7 [online only]; Fig. 4). Native bee richness had three additional competing models, the first including floral diversity ($\chi^2 = 7.00$, $P = 0.008$), the second including percent impervious surface ($\chi^2 = 7.07$, $P = 0.008$), and the third including percent impervious surface ($\chi^2 = 3.21$, $P = 0.073$), floral diversity ($\chi^2 = 4.90$, $P = 0.023$), and their interaction ($\chi^2 = 1.11$, $P = 0.29$).

3) *How do landscape-scale impervious surface and local floral resources influence wild bee functional traits?*

The relationship between the richness of bees possessing each functional trait and our environmental predictors was variable (Fig. 3) with many competing models for most of our functional trait response variables. However, floral diversity did appear to influence the richness of bees for some functional traits. The richness of above-ground nesting bees, oligolectic bees, parasitic bees, and noneusocial bees were best supported by null models (Table 4; Supp Table S7 [online only]; Figs. 5), but most had at least one other competing model including floral diversity or percent impervious surface. Specifically, the richness of above-ground nesting bees had two competing models: one including percent impervious surface ($\chi^2 = 0.94$, $P = 0.33$) and the other including floral diversity ($\chi^2 = 2.05$, $P = 0.15$). Oligolectic bees had one competing model including percent impervious surface ($\chi^2 = 2.19$, $P = 0.14$). Noneusocial bees had two competing models: one including floral diversity ($\chi^2 = 1.58$, $P = 0.21$) and one including percent impervious surface ($\chi^2 = 0.93$, $P = 0.34$). The richness of ground nesting, polylectic, and eusocial bees were best supported by models including only floral diversity, and in all cases, we observed positive relationships (Table 4; Fig. 5; Supp Table S7 [online only]). Additionally, the richness of ground nesting, polylectic, and eusocial bees had competing null models, and eusocial bees had an additional competing model that included percent impervious surface ($\chi^2 = 4.65$, $P = 0.031$), floral diversity ($\chi^2 = 6.79$, $P = 0.009$) and their interaction ($\chi^2 = 4.38$, $P = 0.03$). Lastly, bee size at each sampling period was best supported by a null model (Table 4; Supp Table S7 [online only]; Figs. 5).

Discussion

Urbanization continues to grow rapidly worldwide with major implications for a variety of organisms (McKinney 2002, 2006, 2008)—yet we are still only beginning to understand how specific landscape features affect different organisms within cities and surrounding suburban areas. We investigated how bee communities respond to increasing urbanization along a gradient in the Chicagoland region by investigating both landscape-scale and local-scale variables. We find that landscape-scale impervious surface and the diversity of local floral resources give rise to variation in bee communities along the urbanization gradient. Impervious surface influenced bee community composition, with highly urban areas being composed of more non-native bee species and fewer native bee species. Additionally, local floral diversity emerged as an important predictor of total bee richness, native bee richness, and the richness of polylectic, ground nesting, and generalist bee species independent

Table 3. Model selection results for the effects of % impervious surface (500 m) and floral diversity (Shannon–Weiner index) on bee community composition metrics (abundance, species richness, evenness, diversity [Shannon–Wiener index], Bray–Curtis dissimilarity)

Response (distribution)	Model	K	Δ AICc	Relative likelihood	Weight
Bee abundance (negative binomial)	Null	1	0.00	1.00	0.34
	% impervious surface × floral diversity	4	0.81	0.67	0.23
	Floral diversity	2	1.54	0.46	0.16
	% impervious surface	2	1.57	0.46	0.16
	% impervious surface + floral diversity	3	2.18	0.34	0.11
Bee richness (Poisson)	Floral diversity	2	0.00	1.00	0.54
	% impervious surface × floral diversity	4	2.11	0.35	0.19
	% impervious surface + floral diversity	3	2.52	0.28	0.16
	Null	1	3.73	0.15	0.084
	% impervious surface	2	5.57	0.062	0.033
Bee evenness	Null	1	0.00	1.00	0.98
	Floral diversity	2	7.48	0.024	0.023
	% impervious surface	2	12.66	0.0017	0.0017
	% impervious surface + floral diversity	3	20.37	0.000038	3.7e-5
	% impervious surface × floral diversity	4	33.02	6.76e-8	6.59e-8
Bee diversity (α)	Null	1	0.00	1.00	0.79
	Floral diversity	2	2.75	0.25	0.20
	% impervious surface	2	10.44	0.0054	0.0043
	% impervious surface + floral diversity	3	13.71	0.0011	0.00084
	% impervious surface × floral diversity	4	24.89	3.9e-6	7.0e-8
Bray–Curtis	Null	1	0.00	1.00	0.99
	% impervious surface	2	9.08	0.011	0.011

The best fit models are in bold.

of impervious surface. Taken together, these results indicate that urban areas can support diverse bee communities, but these communities do differ markedly in terms of species composition, and in particular, whether the species are native or not native to North America. Furthermore, local floral diversity is an important variable that may compensate for some of the negative effects of impervious surface in urban areas.

Variation in bee community composition along the urbanization gradient was complex, with bee abundance, evenness, and diversity, remaining relatively stable, but composition differing between urban and suburban sites overall. The changes in community composition along the urbanization gradient that we observed appear to be largely driven by the richness of native and non-native species. Non-native bees responded positively to landscape-scale impervious surface with the greatest richness of non-native bees located at sites in the urban core of Chicago (Fig. 4). These results are consistent with the emerging hypothesis that urban areas support many non-native bee species (Matteson et al. 2008, Makinson et al. 2017, Fitch et al. 2019, Wilson and Jamieson 2019). Some non-native bee species have been documented exclusively in urban environments, possibly due to a high abundance of non-native plants (Russo 2016, Makinson et al. 2017, Portman et al. 2019, Gruver and CaraDonna 2020). There is some evidence that non-native plants may facilitate the spread of non-native bee species (O'Brien et al. 2012, Gibbs and Dathe 2017, Gibbs et al. 2017, Martins et al. 2017), but others have suggested that non-native plants do not facilitate the spread of non-native bees (Fitch et al. 2019). We found that non-native plants were the most abundant and appeared to be the main floral resources for bees at all study sites, with native and non-native bees visiting non-native plants 77% and 88% of the time respectively. In addition, we found a relationship between urbanization and the proportion of non-native inflorescences, with highly urban areas having a higher proportional abundance and richness of non-native inflorescences. This pattern suggests non-native plants may play a partial role in supporting non-native bees in highly urban areas.

Interestingly, four of the thirteen non-native species found along our gradient were found exclusively at urban sites, three of which are oligolectic *Pseudoanthidium nanum* (Mocsáry) (Megachilidae), *Chelostoma rapunculi* (Lepeletier) (Megachilidae) or have a strong association with a specific group of plants (*Megachile apicalis*). For example, both *M. apicalis* and *P. nanum* are commonly found visiting plants in the *Centaurea* genus (knapweeds, Asteraceae) (Müller and Bansac 2004, Gonzalez et al. 2017). *Centaurea stoebe* (L.) (spotted knapweed) is an invasive knapweed in Illinois and was only present at the two most urban sites where both *M. apicalis* and *P. nanum* were documented. This suggests that in urban areas, especially those that are unmanaged, specific invasive or non-native plants may create suitable habitat for non-native bees who have strong associations with them.

In contrast to the patterns we observed with non-native bees, native bees negatively responded to landscape-scale impervious surface, but positively responded to local floral diversity. Not only did sites with high local floral diversity support more native bee species, but they also supported a greater number of bee species overall, regardless of whether the species was native or non-native. Local floral resources have been shown to positively affect bee species richness in urban areas, and have been suggested to be one of the most important and generally consistent factors to predict bee diversity and population sizes (Fründ et al. 2010, Roulston and Goodell 2011, Tonietto et al. 2011, Pardee and Philpott 2014, Wilson and Jamieson 2019). Habitats with greater floral diversity provide bees with a wider array of food resources to fulfill their foraging requirements and provision for their offspring (Roulston and Goodell 2011, Winfree et al. 2011). Our results showing a positive effect of floral diversity on bee richness suggest that sites with diverse floral resources may be able to help sustain diverse bee communities in urban areas even if there is little surrounding greenspace, substantiating some claims that urban areas may be refuges for bees (Hall et al. 2017). Although we do see more non-native bees in the urban core of Chicago, providing habitats in cities with high floral diversity may promote more bee species, including native bees.

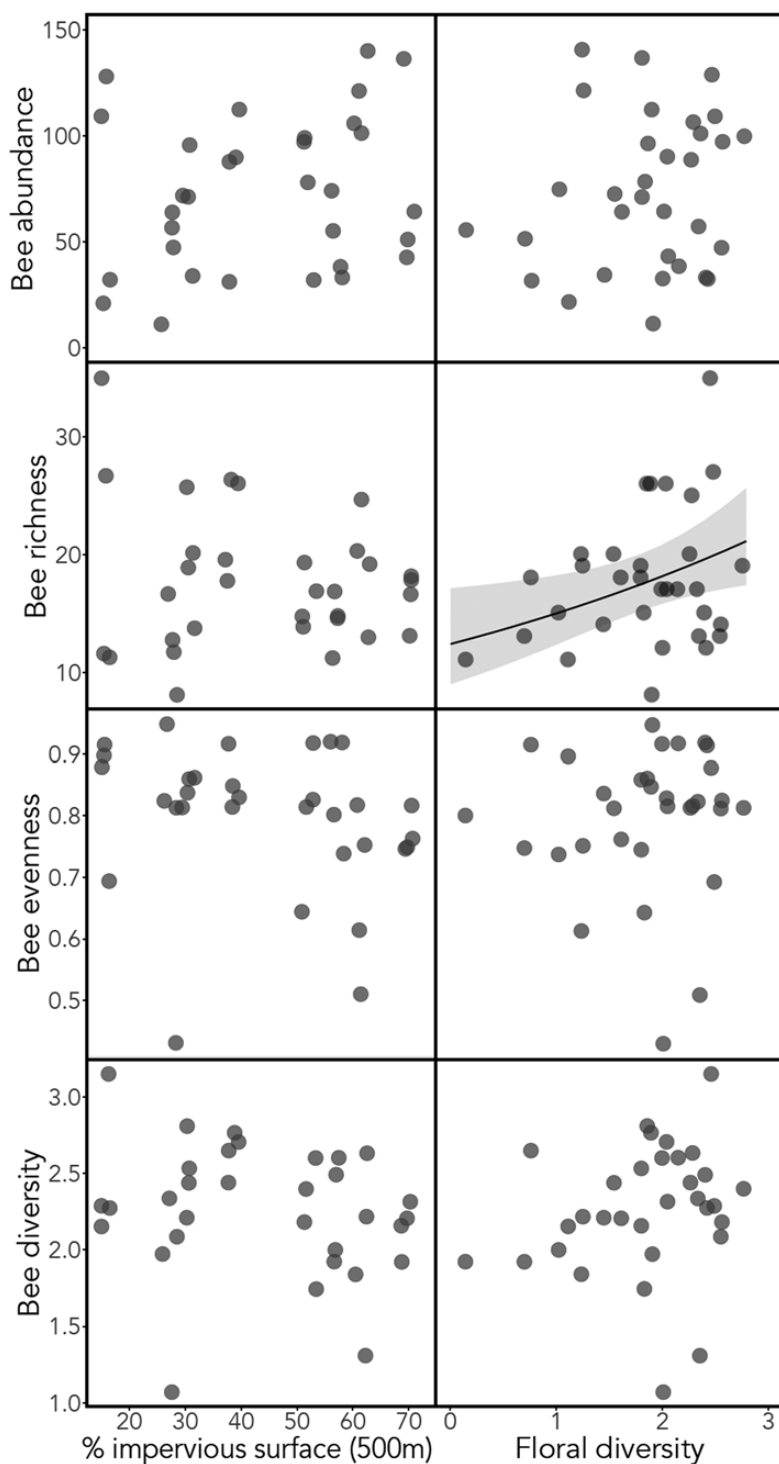


Fig. 3. The effect of % impervious surface and floral diversity (Shannon–Wiener index) on bee metrics: abundance, species richness, evenness, diversity (Shannon–Wiener index). Bee species richness was positively related to floral diversity. Dots represent sampling periods.

Bee functional traits had variable responses to our landscape predictors, but overall functional traits did not respond strongly to landscape-scale impervious surface along the urbanization gradient, suggesting that impervious surface may not act as a strong filter on the functional traits examined. Instead, some functional traits (ground nesting, eusocial, and polylectic bees) responded positively to local floral diversity, suggesting that local floral diversity may

support more ground nesting, eusocial, and polylectic bee species. The richness of ground nesting bee species did not respond to impervious surface but responded positively to local floral diversity across the urbanization gradient. It has been suggested that abundant floral resources may be an indicator of the abundance of nesting resources (Roulston and Goodell 2011). However, there is still much to learn regarding the role nesting resources play in supporting bees

Table 4. Model selection results for the effects of % impervious surface (500 m) and floral diversity (Shannon–Wiener index) on the species richness of bees representing each functional trait and the average intertegular distance of bees

Response (distribution)	Model	K	ΔAIC_c	Relative likelihood	Weight
Richness of above-ground nesting bees (Poisson)	Null	1	0.00	1.00	0.37
	Floral diversity	2	0.28	0.87	0.32
	% impervious surface	2	1.57	0.46	0.17
	% impervious surface + floral diversity	3	2.59	0.27	0.10
	% impervious surface $\times$ floral diversity	4	4.71	0.095	0.035
Richness of ground nesting bees (Poisson)	Floral diversity	2	0.00	1.00	0.44
	Null	1	1.04	0.59	0.26
	% impervious surface + floral diversity	3	2.61	0.27	0.12
	% impervious surface $\times$ floral diversity	4	3.04	0.22	0.096
	% impervious surface	2	3.27	0.19	0.085
Richness of polylectic bees (negative binomial)	Floral diversity	2	0.00	1.00	0.43
	Null	1	1.24	0.54	0.23
	% impervious surface	2	2.54	0.28	0.12
	% impervious surface + floral diversity	3	2.50	0.29	0.12
	% impervious surface $\times$ floral diversity	4	3.12	0.21	0.091
Richness of oligolectic bees (Poisson)	Null	1	0.00	1.00	0.43
	% impervious surface	2	1.04	0.59	0.28
	Floral diversity	2	2.37	0.31	0.15
	% impervious surface + floral diversity	3	3.67	0.16	0.076
	% impervious surface $\times$ floral diversity	4	6.37	0.041	0.020
Richness of parasitic bees (Poisson)	Null	1	0.00	1.00	0.71
	% impervious surface	2	3.079	0.21	0.15
	Floral diversity	2	3.77	0.15	0.83
	% impervious surface + floral diversity	3	6.55	0.0378	0.027
	% impervious surface $\times$ floral diversity	4	20.13	0.000043	3.0e-5
Richness of eusocial bees (Poisson)	Floral diversity	2	0.00	1.00	0.39
	% impervious surface $\times$ floral diversity	4	0.61	0.74	0.29
	Null	1	1.89	0.39	0.15
	% impervious surface + floral diversity	3	2.20	0.33	0.13
	% impervious surface	2	4.32	0.12	0.045
Richness of noneusocial bees (Poisson)	Null	1	0.00	1.00	0.41
	Floral diversity	2	0.82	0.66	0.28
	% impervious surface	2	1.58	0.45	0.19
	% impervious surface + floral diversity	3	2.98	0.22	0.093
	% impervious surface $\times$ floral diversity	4	5.26	0.072	0.030
Richness of native bees (negative binomial)	% impervious surface + floral diversity	3	0.00	1.00	0.34
	Floral diversity	2	0.0087	0.99	0.33
	% impervious surface	2	1.47	0.48	0.16
	% impervious surface $\times$ floral diversity	4	1.98	0.37	0.12
	Null	1	3.97	0.14	0.046
Richness of non-native bees (Poisson)	% impervious surface	2	0.00	1.00	0.68
	% impervious surface $\times$ floral diversity	4	5.34	0.069	0.047
	% impervious surface + floral diversity	3	2.52	0.28	0.19
	Null	1	5.39	0.067	0.046
	Floral diversity	2	7.72	0.021	0.014
Average intertegular distance (mm)	Null	1	0.00	1.00	0.92
	Floral diversity	2	5.04	0.080	0.074
	% impervious surface	2	10.80	0.0045	0.0042
	% impervious surface + floral diversity	3	16.08	0.00032	0.00030
	% impervious surface $\times$ floral diversity	4	22.94	0.000010	9.62e-5

Best fit models are represented in bold.

in urban habitats, and the nesting requirements and basic biology for many bees still remain understudied (Roulston and Goodell 2011, Harmon-Threatt 2020). An in-depth investigation of nesting resources in urban areas will help to elucidate the role nesting resources play in supporting bees in urban habitats and will help to explain the patterns we observed in bee nesting behavior along the urbanization gradient.

The richness of eusocial, noneusocial, and parasitic bees did not respond to impervious surface along the urbanization gradient.

Furthermore, eusocial bee species responded positively to local floral diversity, but noneusocial and parasitic bees did not respond. The high richness of eusocial bees at sites with high floral diversity may be due to these sites providing continuous and abundant floral resources with different species blooming across the active season (Hemberger et al. 2020). This is especially important for eusocial bee species who have a long foraging period that spans most of the season, as opposed to some noneusocial species who have shorter foraging periods (Michener 2007, Roulston and Goodell 2011). In addition, many

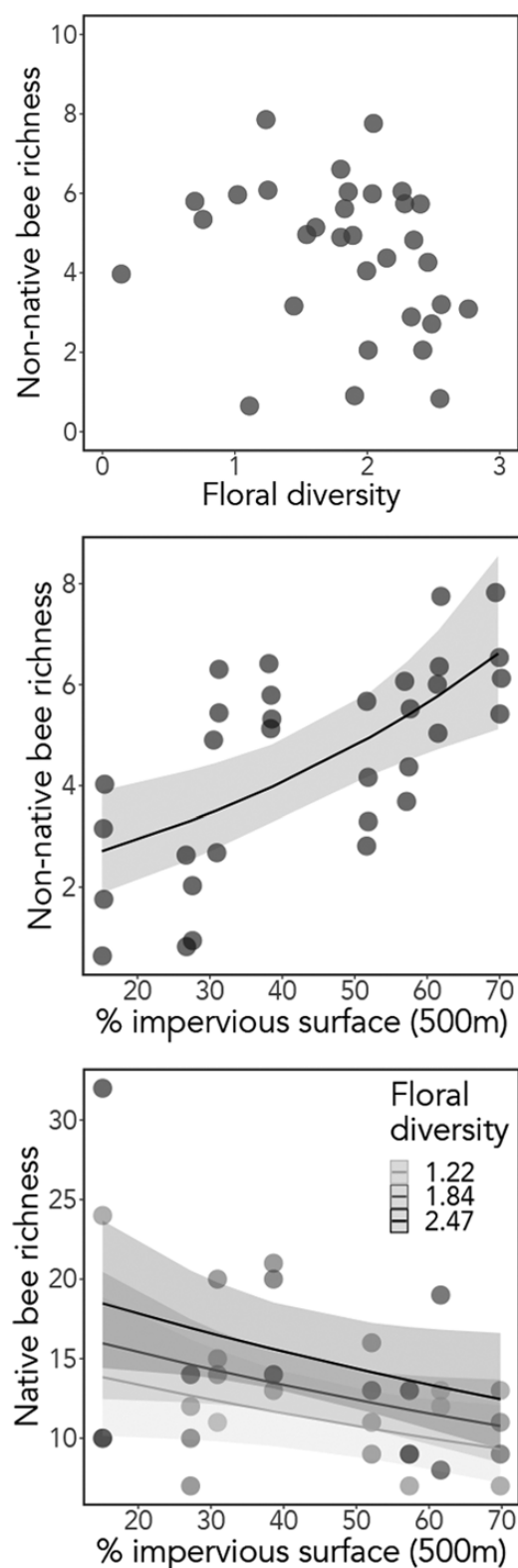


Fig. 4. The effect of % impervious surface (500 m) and floral diversity (Shannon-Wiener index) on non-native and native bee species richness. The richness of non-native bee species was positively related to % impervious surface, but not floral diversity. The richness of native bee species was negatively related to % impervious surface and positively related to floral diversity. Dots represent one sampling period and are colored on a continuous scale according to floral diversity. Lines represent three levels of floral diversity have been selected to illustrate trends as % impervious surface increases.

eusocial bee species, such as bumble bees and honey bees, are able to forage longer distances; thus, in disturbed landscapes, some eusocial bees may travel to areas with higher floral diversity and more abundant floral resources (Carvell et al. 2012, Redhead et al. 2016).

Although parasitic species richness did not appear to respond to urbanization or floral diversity along the gradient, parasitic bees were observed in higher abundance in suburban sites than urban sites, with 16 individuals found in suburban sites and only four found in urban sites. Parasitic species rely on their hosts for survival and it has been suggested that parasitic bees are a sensitive indicator of bee community stability and health (Marcogliese 2004, Hudson et al. 2006, Sheffield et al. 2013). The higher abundance of parasites at suburban sites may therefore suggest suburban sites are able to maintain more diverse bee communities than urban sites, but more work is needed to support this hypothesis.

Lastly, the richness of polylectic and oligolectic bees did not respond to impervious surface, but polylectic bees did positively respond to local floral diversity along the urbanization gradient. Oligolectic bees were somewhat rare along our urbanization gradient. We documented a total of 164 oligolectic individuals (6.5% of all bees) from eight species (10% of all bees), three of which were non-native species. Other studies have also detected very few oligolectic bees in highly urban areas, possibly due to a urban areas lacking the specific floral resources they require (Cane et al. 2006, Winfree et al. 2011). On the other hand, polylectic bees were much more common, and we observed a greater species richness at sites with high floral diversity. Floral diverse sites along our urbanization gradient may provide diverse resources and balanced nutrients that are important for polylectic species who need to visit a variety of plants to fulfill their dietary requirements.

While we are beginning to better understand how urban areas affect wild bee communities, future work should focus on broadening urban bee studies to explore the interplay between landscape features, bee communities, and socioeconomic factors (e.g., Lowenstein et al. 2014). Urban areas are heterogeneous landscapes, with different landscape features and vegetation components that can interplay to shape pollinator communities (Matteson et al. 2013). Our field sites were generally unmanaged areas surrounding train tarmacs, while managed gardens and large urban parks and greenspaces could have varying effects on bee communities. In addition, our field sites only encompassed a small portion of Chicago and the surrounding suburbs. Yet, Chicago is a large and diverse city with many different neighborhoods that differ in landscape features and modifications. Furthermore, social inequalities can strongly shape landscape features and may therefore underlie many ecological patterns in cities (Schell et al. 2020). Thus, it is critical that future studies explore ecological communities across neighborhoods and cities of differing socioeconomic status in order to more broadly understand the interplay among social inequalities and urban biodiversity.

Across an urbanization gradient in the Chicagoland region, we documented 2,331 bees comprised of 83 species from five families. The bees represent a diversity of traits and life history strategies including (but not limited to) non-native species, ground nesters, specialists, cleptoparasites, and an endangered bumble bee species. However, our study was conducted over one year from June to August; this sampling may miss natural year-to-year turnover in bee (and plant) communities and may exclude some early season foraging bee species (such as early season species Andrenidae). Multi-year, full-season study of bee communities in urban areas are generally lacking and are needed to better understand how urban areas shape bee communities. Nevertheless, our study still provides a solid foundation for understanding how urbanization

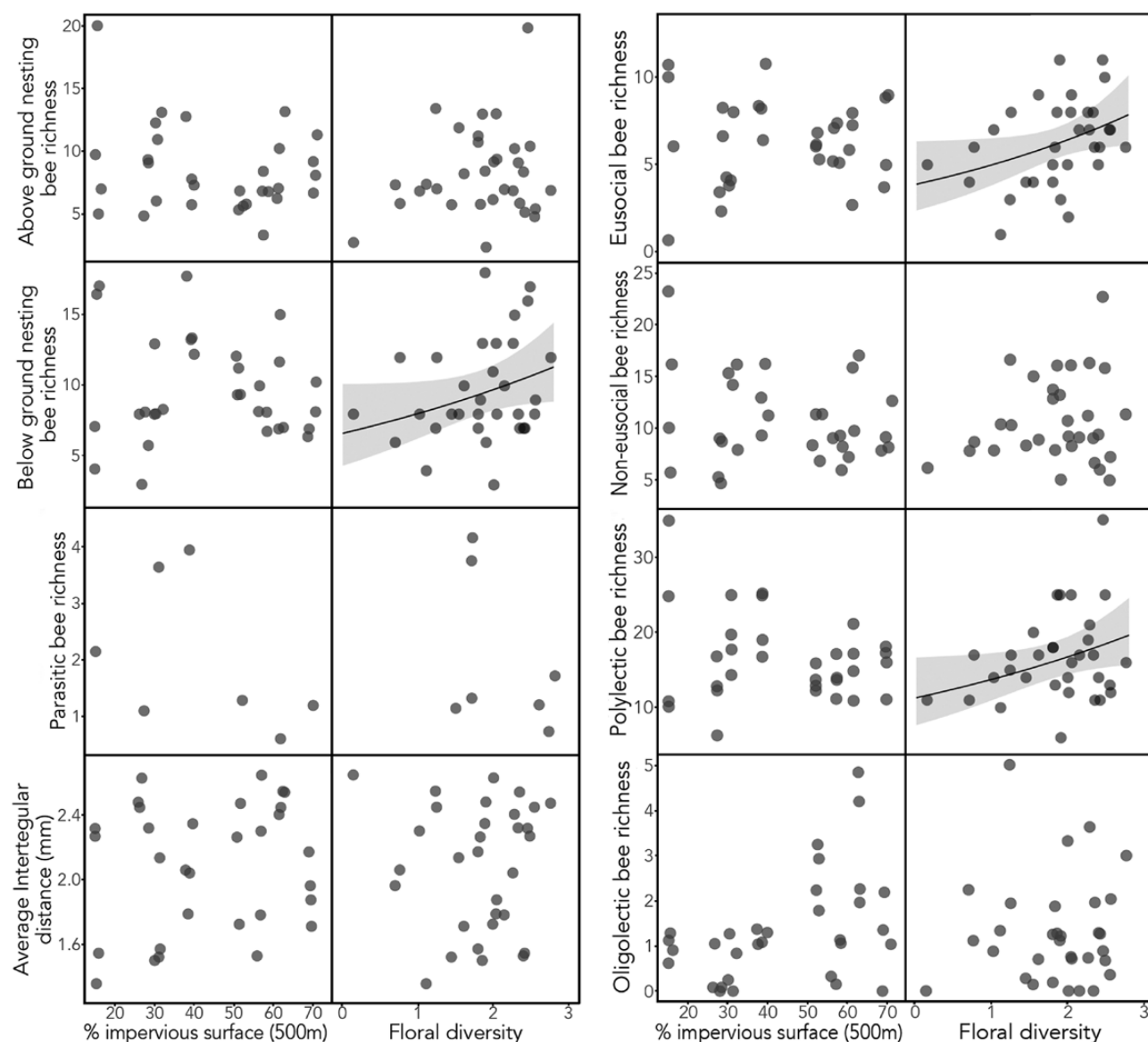


Fig. 5. The effect of % impervious surface (500 m) and floral diversity (Shannon–Wiener index) on the richness of bees possessing each functional trait. There were positive relationships between floral diversity and the richness of below ground nesting bees, polylectic bees and eusocial bees, with all three traits having the greatest richness of bees at sites with high floral diversity. Dots represent sampling periods.

and floral resources shape bee communities in our study region. Notably, our study adds to the growing body of literature suggesting that highly urban areas support more non-native bee species and fewer native bee species. Urban areas have been suggested to be refuges for bees (Hall et al. 2017), but it is important we understand the bee communities our urban areas are supporting. Critically, urbanization may pose as a threat to native bees, but providing diverse floral communities in urban areas can help to support native bees in cities and should be a priority for their conservation. Additionally, it is crucial to engage and educate the public on the importance of native bees and bee diversity to better support and promote biodiversity in cities.

Supplementary Data

Supplementary data are available at *Environmental Entomology* online.

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References Cited

- Anderson, M. J. 2017. Permutational multivariate analysis of variance (PERMANOVA), pp. 1–15. *In* Wiley StatsRef Stat. Ref. Online. American Cancer Society.
- Ascher, J. S., and J. Pickering. 2020. Discover life bee species guide and world checklist (Hymenoptera: Apoidea: Anthophila). Available from <https://www.discoverlife.org/mp/20q?search=Apoidea>.

- Banaszak-Cibicka, W., and M. Żmihorski. 2012. Wild bees along an urban gradient: winners and losers. *J. Insect Conserv.* 16: 331–343.
- Bates, D., M. Maechler, B. Bolker, and S. Walker. 2015. Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* 67: 1–48.
- Biesmeijer, J. C., S. P. Roberts, M. Reemer, R. Ohlemüller, M. Edwards, T. Peeters, A. P. Schaffers, S. G. Potts, R. Kleukers, C. D. Thomas, *et al.* 2006. Parallel declines in pollinators and insect-pollinated plants in Britain and the Netherlands. *Science*. 313: 351–354.
- Brooks, M., K. Kristensen, K. van Benthem, A. Magnusson, C. Berg, A. Nielsen, M. Maechler, and B. Bolker. 2017. glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. *R J.* 9: 378–400.
- Burkle, L. A., J. C. Marlin, and T. M. Knight. 2013. Plant-pollinator interactions over 120 years: loss of species, co-occurrence, and function. *Science*. 339: 1611–1615.
- Cane, J. H., R. L. Minckley, L. J. Kervin, T. H. Roulston, and N. M. Williams. 2006. Complex responses within a desert bee guild (Hymenoptera: Apoidea) to urban habitat fragmentation. *Ecol. Appl.* 16: 632–644.
- Cardoso, M. C., and R. B. Gonçalves. 2018. Reduction by half: the impact on bees of 34 years of urbanization. *Urban Ecosyst.* 21: 943–949.
- Carvell, C., W. C. Jordan, A. F. G. Bourke, R. Pickles, J. W. Redhead, and M. S. Heard. 2012. Molecular and spatial analyses reveal links between colony-specific foraging distance and landscape-level resource availability in two bumblebee species. *Oikos*. 121: 734–742.
- Choate, B. A., P. L. Hickman, and E. A. Moretti. 2018. Wild bee species abundance and richness across an urban–rural gradient. *J. Insect Conserv.* 22: 391–403.
- Czech, B., P. R. Krausman, and P. K. Devers. 2000. Economic associations among causes of species endangerment in the United States. *BioScience*. 50: 593–601.
- Droege, S. 2015. The very handy manual how to catch and identify bees and manage a collection. USGS Native Bee Inventory and Monitoring Lab, Beltsville, Maryland, USA.
- Fitch, G., C. J. Wilson, P. Glaum, C. Vaidya, M. C. Simao, and M. A. Jamieson. 2019. Does urbanization favour exotic bee species? Implications for the conservation of native bees in cities. *Biol. Lett.* 15: 20190574.
- Fortel, L., M. Henry, L. Guilhaud, A. L. Guirao, M. Kuhlmann, H. Mouret, O. Rollin, and B. E. Vaissière. 2014. Decreasing abundance, increasing diversity and changing structure of the wild bee community (Hymenoptera: Anthophila) along an urbanization gradient. *PLoS One*. 9: e104679.
- Fründ, J., K. E. Linsenmair, and N. Blüthgen. 2010. Pollinator diversity and specialization in relation to flower diversity. *Oikos*. 119: 1581–1590.
- Garibaldi, L. A., I. Steffan-Dewenter, R. Winfree, M. A. Aizen, R. Bommarco, S. A. Cunningham, C. Kremen, L. G. Carvalheiro, L. D. Harder, O. Afik, *et al.* 2013. Wild pollinators enhance fruit set of crops regardless of honey bee abundance. *Science*. 339: 1608–1611.
- Gibbs, J., and H. H. Dathe. 2017. First records of *Hylaeus (Paraprosopis) pictipes* Nylander, 1852 (Hymenoptera: Colletidae) in North America. *Check List*. 13: 2116.
- Gibbs, J., J. S. Ascher, M. G. Rightmyer, and R. Isaacs. 2017. The bees of Michigan (Hymenoptera: Apoidea: Anthophila), with notes on distribution, taxonomy, pollination, and natural history. *Zootaxa*. 4352: 1–160.
- Gonzalez, V. H., A. Olsen, M. Mallula, A. Tosunoglu, I. Çakmak, J. Hranitz, and J. Barthell. 2017. Bee visitors of *Centaurea solstitialis* L. (Asteraceae) in an urban environment in northwestern Turkey. *Arthropod Plant Interact.* 11: 403–409.
- Goulson, D., E. Nicholls, C. Botías, and E. L. Rotheray. 2015. Bee declines driven by combined stress from parasites, pesticides, and lack of flowers. *Science*. 347: 1255957.
- Gruver, A., and P. CaraDonna. 2020. First records of *Megachile apicalis* (Hymenoptera: Megachilidae) for Illinois found in heavily urbanized areas within the city of Chicago. *Gt. Lakes Entomol.* 53: 41–46.
- Hall, D. M., G. R. Camilo, R. K. Tonietto, J. Ollerton, K. Ahrné, M. Arduser, J. S. Ascher, K. C. R. Baldock, R. Fowler, G. Frankie, *et al.* 2017. The city as a refuge for insect pollinators: insect pollinators. *Conserv. Biol.* 31: 24–29.
- Harmon-Threalt, A. 2020. Influence of nesting characteristics on health of wild bee communities. *Annu. Rev. Entomol.* 65: 39–56.
- Hemberger, J., A. Frappa, G. Witynski, and C. Gratton. 2020. Saved by the pulse? Separating the effects of total and temporal food abundance on the growth and reproduction of bumble bee microcolonies. *Basic Appl. Ecol.* 45: 1–11.
- Hsieh, T. C., K. H. Ma, and A. Chao. 2016. iNEXT: an R package for rarefaction and extrapolation of species diversity (Hill numbers). *Methods Ecol. Evol.* 7: 1451–1456.
- Hudson, P. J., A. P. Dobson, and K. D. Lafferty. 2006. Is a healthy ecosystem one that is rich in parasites? *Trends Ecol. Evol.* 21: 381–385.
- Kindt, R., and R. Coe. 2005. Tree diversity analysis: a manual and software for common statistical methods for ecological and biodiversity studies. World Agroforestry Centre, Nairobi, Kenya.
- Lowe, A. D., and J. L. Foltz-Sweat. 2017. Effect of floral diversity and urbanization on bee species community composition in Phoenix, Arizona. *J. Ariz.-Nev. Acad. Sci.* 47: 6–18.
- Lowenstein, D. M., K. C. Matteson, I. Xiao, A. M. Silva, and E. S. Minor. 2014. Humans, bees, and pollination services in the city: the case of Chicago, IL (USA). *Biodivers. Conserv.* 23: 2857–2874.
- Makinson, J. C., C. G. Threlfall, and T. Latty. 2017. Bee-friendly community gardens: impact of environmental variables on the richness and abundance of exotic and native bees. *Urban Ecosyst.* 20: 463–476.
- Marcogliese, D. J. 2004. Parasites: small players with crucial roles in the ecological theater. *EcoHealth*. 1: 151–164.
- Martins, K. T., É. Normandin, and J. S. Ascher. 2017. *Hylaeus communis* (Hymenoptera: Colletidae), a new exotic bee for North America with generalist foraging and habitat preferences. *Can. Entomol.* 149: 377–390.
- Matteson, K. C., and G. A. Langellotto. 2010. Determinates of inner city butterfly and bee species richness. *Urban Ecosyst.* 13: 333–347.
- Matteson, K. C., J. S. Ascher, and G. A. Langellotto. 2008. Bee richness and abundance in New York City urban gardens. *Ann. Entomol. Soc. Am.* 101: 140–150.
- Matteson, K. C., J. B. Grace, and E. S. Minor. 2013. Direct and indirect effects of land use on floral resources and flower-visiting insects across an urban landscape. *Oikos*. 122: 682–694.
- McGill, B. J., B. J. Enquist, E. Weiher, and M. Westoby. 2006. Rebuilding community ecology from functional traits. *Trends Ecol. Evol.* 21: 178–185.
- McKinney, M. L. 2002. Urbanization, biodiversity, and conservation. *BioScience*. 52: 883.
- McKinney, M. L. 2006. Urbanization as a major cause of biotic homogenization. *Biol. Conserv.* 127: 247–260.
- McKinney, M. L. 2008. Effects of urbanization on species richness: a review of plants and animals. *Urban Ecosyst.* 11: 161–176.
- Michener, C. 2007. The bees of the world, 2nd ed. John Hopkins University Press, Baltimore and London.
- Michener, C. D., R. J. McGinley, and B. N. Danforth. 1994. The bee genera of North and Central America (Hymenoptera: Apoidea). Smithsonian Institution Press, Washington, DC.
- Mitchell, T. B. 1962. Bees of the Eastern United States volume II. Tech. Bull. N. C. Agric. Exp. States, Raleigh, North Carolina, USA.
- Müller, A., and N. Bansac. 2004. A specialized pollen-harvesting device in western palaearctic bees of the genus *Megachile* (Hymenoptera, Apoidea, Megachilidae). *Apidologie*. 35: 329–337.
- O'Brien, M. F., D. R. Swanson, and J. Monsma. 2012. *Anthidium Oblongatum* (Apoidea: Megachilidae) confirmed as a Michigan resident, with notes on other Michigan *Anthidium* species. *Gt. Lakes Entomol.* 45: 102–105.
- Oksanen, J., G. Blanchet, M. Friendly, R. Kindt, L. Pierre, D. McGlinn, P. R. Minchin, R. B. O'Hara, G. L. Simpson, P. Solymos, *et al.* 2019. Vegan: community ecology package. R package version 2.5–6. <http://vegan.r-forge.r-project.org>.
- Pardee, G. L., and S. M. Philpott. 2014. Native plants are the bee's knees: local and landscape predictors of bee richness and abundance in backyard gardens. *Urban Ecosyst.* 17: 641–659.
- Portman, Z. M., S. J. Burrows, T. Griswold, M. Arduser, A. J. Irber, R. K. Tonietto, and D. P. Cariveau. 2019. First records of the adventive *Pseudoanthidium nanum* (Mocsáry) (Hymenoptera: Megachilidae) in

- Illinois and Minnesota, with notes on its identification and taxonomy. *Gt. Lakes Entomol.* 52: 12–20.
- Potts, S. G., J. C. Biesmeijer, C. Kremen, P. Neumann, O. Schweiger, and W. E. Kunin. 2010. Global pollinator declines: trends, impacts and drivers. *Trends Ecol. Evol.* 25: 345–353.
- Redhead, J. W., S. Dreier, A. F. Bourke, M. S. Heard, W. C. Jordan, S. Sumner, J. Wang, and C. Carvell. 2016. Effects of habitat composition and landscape structure on worker foraging distances of five bumble bee species. *Ecol. Appl.* 26: 726–739.
- Roulston, T. H., and K. Goodell. 2011. The role of resources and risks in regulating wild bee populations. *Annu. Rev. Entomol.* 56: 293–312.
- Russo, L. 2016. Positive and negative impacts of non-native bee species around the World. *Insects.* 7: 69.
- Schell, C. J., K. Dyson, T. L. Fuentes, S. D. Roches, N. C. Harris, D. S. Miller, C. A. Woelfle-Erskine, and M. R. Lambert. 2020. The ecological and evolutionary consequences of systemic racism in urban environments. *Science.* 369:eaay4497.
- Sheffield, C. S., A. Pindar, L. Packer, and P. G. Kevan. 2013. The potential of cleptoparasitic bees as indicator taxa for assessing bee communities. *Apidologie.* 44: 501–510.
- Sukopp, H., and P. Werner. 1982. Nature in cities: a report and review of studies and experiments concerning ecology, wildlife, and nature conservation in urban and suburban areas. Council of Europe, Strasbourg, France.
- Tonietto, R., J. Fant, J. Ascher, K. Ellis, and D. Larkin. 2011. A comparison of bee communities of Chicago green roofs, parks and prairies. *Landsc. Urban Plan.* 103: 102–108.
- Wenzel, A., I. Grass, V. V. Belavadi, and T. Tschardt. 2020. How urbanization is driving pollinator diversity and pollination – a systematic review. *Biol. Conserv.* 241: 108321.
- Williams, P. H., R. W. Thorp, L. L. Richardson, and S. R. Colla. 2014. Bumble bees of North America: an identification guide. Princeton University Press, New Jersey.
- Wilson, C. J., and M. A. Jamieson. 2019. The effects of urbanization on bee communities depends on floral resource availability and bee functional traits. *PLoS One.* 14: e0225852.
- Winfree, R., N. M. Williams, J. Dushoff, and C. Kremen. 2007. Native bees provide insurance against ongoing honey bee losses. *Ecol. Lett.* 10: 1105–1113.
- Winfree, R., I. Bartomeus, and D. P. Cariveau. 2011. Native pollinators in anthropogenic habitats. *Annu. Rev. Ecol. Evol. Syst.* 42: 1–22.